

LANDVERIFYCM

DIGITAL LAND VERIFICATION FOR SAFER PROPERTY TRANSACTIONS

PKFokam Institute of Excellence | Computing & Software Engineering | Capstone 2026

SECURE • BILINGUAL
WEB + MOBILE

01 PROBLEM PROPOSAL

Why land buyers need a faster, safer way to verify ownership.

Land fraud in Cameroon exposes citizens and investors to forged records, conflicting claims, costly disputes, and loss of property.

- **Duplicate or forged titles** — the same identity may be reused.
- **Hidden parcel overlap** — GPS conflicts are difficult to detect manually.
- **Fragmented evidence** — ownership and documents are hard to confirm quickly.
- **Low buyer confidence** — decisions are made without timely, reliable facts.

CORE NEED One trusted registry that verifies the title, location, and ownership trail before a transaction.

02 IMPLEMENTATION METHODOLOGY

A secure, data-driven workflow from record entry to verification result.

- 1 Design & requirements** — map buyer/admin journeys and deliver English–French access.
- 2 Build the platform** — React web + Expo mobile → Express REST API → PostgreSQL/Prisma.
- 3 Automate verification** — check title uniqueness and GPS distance with the Haversine formula.
- 4 Secure & govern** — JWT, bcrypt, role controls, validation, uploads, and timestamped audit logs.

DECISION RULES <10 m = DUPLICATE • <50 m = SUSPICIOUS • no conflict VALID

03 SOLUTION PROPOSAL

LandVerifyCM: a centralized verification platform for Cameroon.

SEARCH → VERIFY → DECIDE WITH CONFIDENCE

- **Instant registry search** — title number, owner name, or parcel ID.
- **Clear risk result** — Valid, Suspicious, or Duplicate status.
- **Map-based evidence** — GPS parcel location and proximity checks.
- **Complete record** — ownership chain and supporting PDF/image documents.
- **Controlled administration** — upload, edit, deactivate, browse, and audit records.

Outcome: greater transparency, earlier fraud detection, and safer land transactions.

04 SCAN THE APP

Scan with your phone camera to open LandVerifyCM in the browser.



SCAN • VERIFY • PROTECT

No installation required • Opens directly in your browser
React • TypeScript • Expo • Node.js • PostgreSQL